

Experiment No. 29

Title

Prolog Program for Forward Chaining

Aim

To write a Prolog program to implement **Forward Chaining**.

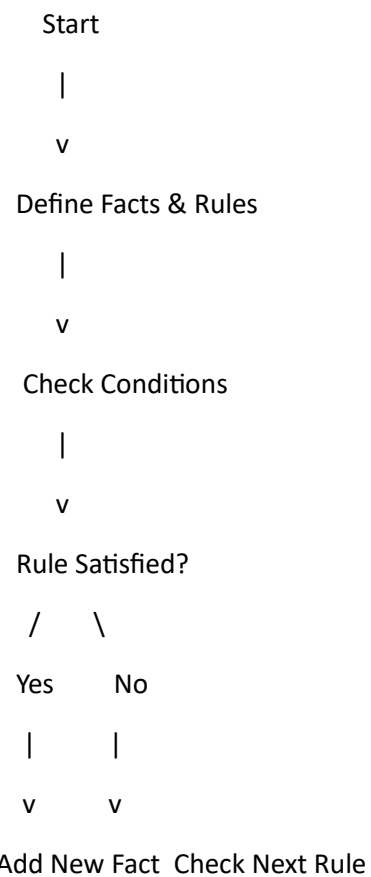
Objective

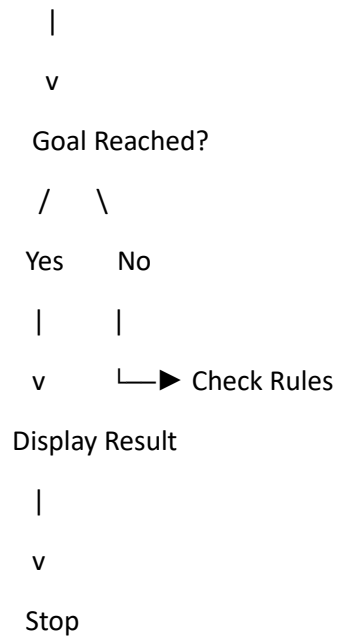
To understand how new facts can be derived from existing facts and rules using forward reasoning.

Algorithm

1. Start.
2. Define the facts and rules.
3. Start with the known facts.
4. Check the conditions of each rule.
5. Derive and add new facts when conditions are satisfied.
6. Continue until the required conclusion is obtained.
7. Stop.

Flowchart





Prolog Program

fact(rain).

rule(rain, wet_ground).

forward_chain(Fact, Result) :-

rule(Fact, Result).

Queries

?- forward_chain(rain, Result).

Sample Output

Result = wet_ground.

Result

The Prolog program successfully implemented **Forward Chaining** and derived a new fact from the given fact and rule.

Conclusion

Thus, Forward Chaining was successfully demonstrated in Prolog by deriving a conclusion from known facts.